

SANAM RAI

Backend Developer · MERN Stack · Kathmandu, Nepal
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SUMMARY

BIT student with hands-on experience building and deploying full-stack MERN applications to production. Built Krishi Bazar — a complete agricultural e-commerce marketplace with JWT authentication, real-time cart management, dual checkout flows, skeleton loading, and a notification system — from scratch and shipped it live. Former QA Intern at Danfe Solution Pvt. Ltd. Strong backend fundamentals: REST API design, JWT auth, real-time systems with Socket.IO, and modular architecture. Self-driven learner with Harvard CS50x and CS50P certifications.

TECHNICAL SKILLS

Backend: Node.js · Express.js · REST API Design · JWT Auth · Socket.IO · bcrypt · Rate Limiting · Morgan · Flask

Frontend: React.js · JavaScript (ES6+) · React Context API · Custom Hooks · React Router · Axios · Tailwind CSS · HTML5 · CSS3

Database: MongoDB (Atlas) · Mongoose · MySQL

Testing & QA: Selenium · Playwright · Pytest · Manual Testing · Functional & Regression Testing

Tools: Git · GitHub · Postman · VS Code · Linux

Deployment: Render · Vercel · MongoDB Atlas

PROJECTS

Krishi Bazar — Agricultural E-Commerce Marketplace — Node.js · Express.js · React.js · MongoDB · JWT · Tailwind CSS · Render · Vercel [\[GitHub\]](#)

Live: krishi-bazar-alpha.vercel.app

- Built and deployed a complete full-stack agricultural marketplace where farmers sell directly to buyers — with product listing, search, category filtering, sorting, and pagination.
- Implemented dual checkout flows: cart-based checkout and Buy Now (single product direct purchase), both handled by a single order controller with role-aware logic.
- Designed a Context-based state architecture (AuthContext, CartContext, NotificationContext) with custom hooks — shared state across all components without prop drilling.
- Built a custom in-app notification system from scratch — mirroring how production toast libraries work, with auto-dismiss, success/error states, and global availability via Context.
- Implemented skeleton loading screens on all data-heavy pages (HomePage, ProductPage, CartPage, CheckoutPage) for a professional loading experience.
- Secured all cart and checkout routes with JWT middleware and React PrivateRoute — unauthenticated users are redirected to login automatically.
- Configured Axios interceptor to attach JWT token to every protected request automatically — eliminating manual header management across the app.
- Deployed backend on Render and frontend on Vercel with MongoDB Atlas as the production database.

Portfolio System — Feature Toggle Architecture — Node.js · Express.js · MongoDB · JWT [\[GitHub\]](#)

Live: sanam-rai.com.np

- Built a backend API with dynamic feature toggles — JWT auth, in-memory caching, rate limiting, pagination, and logging each independently switchable at runtime.
- Designed in-memory caching to reduce DB load and custom pagination supporting query-based navigation.
- Structured with clean modular architecture: controllers, middleware, routes, and config layers fully separated.

YakTalk — Secure Real-Time Chat Application — Node.js · Express · Socket.IO · React · MongoDB · JWT · Tailwind CSS · Render [\[GitHub\]](#)

Live: yaktalk-chatapp-1.onrender.com

- Built and deployed a full-stack real-time chat app with private messaging, online presence tracking, and JWT authentication.
- Secured WebSocket connections by verifying JWT tokens server-side during Socket.IO handshake — preventing unauthorised real-time access.

- Resolved CORS and cross-domain cookie challenges for separate frontend and backend deployment on Render.

Phone Book Application — Python · Pytest · Regex · JSON [GitHub]

- CLI-based CRUD contact manager with Regex input validation, JSON persistence, and Pytest unit tests.

PROFESSIONAL EXPERIENCE

Danfe Solution Pvt. Ltd. · Quality Assurance Intern

Dec 2025 – Mar 2026 · Kathmandu, Nepal

- Conducted manual functional testing across multiple internal systems, identifying and documenting defects with clear reproduction steps.
- Performed regression and validation testing to ensure application stability after updates.
- Collaborated with developers to debug issues, triage defects, and improve product quality.
- Created technical documentation to streamline QA workflows and internal processes.
- Assisted in client onboarding and product training, improving user adoption.

EDUCATION

Tribhuvan University – Bhaktapur Multiple Campus · Bachelor in Information Technology (BIT)

Completed(Certificate provisionally waiting)

Relevant Coursework: Data Structures & Algorithms · DBMS · Software Engineering · Operating Systems · Computer Networks

CERTIFICATIONS

- [CS50x: Introduction to Computer Science — Harvard University / edX](#)
- [CS50P: Introduction to Programming with Python — Harvard University / edX](#)

KEY STRENGTHS

- Ships complete products — Krishi Bazar and YakTalk are both live, deployed, and fully functional end to end.
- Security-aware backend — implements JWT auth, bcrypt, rate limiting, and protected routes by default.
- Strong systems thinking — independently designed a notification system matching how production toast libraries work internally.
- QA mindset — testing background from professional internship informs how code is written and debugged.
- Self-driven — completed two Harvard certifications independently alongside a full university course load.